

Windows and WSL2

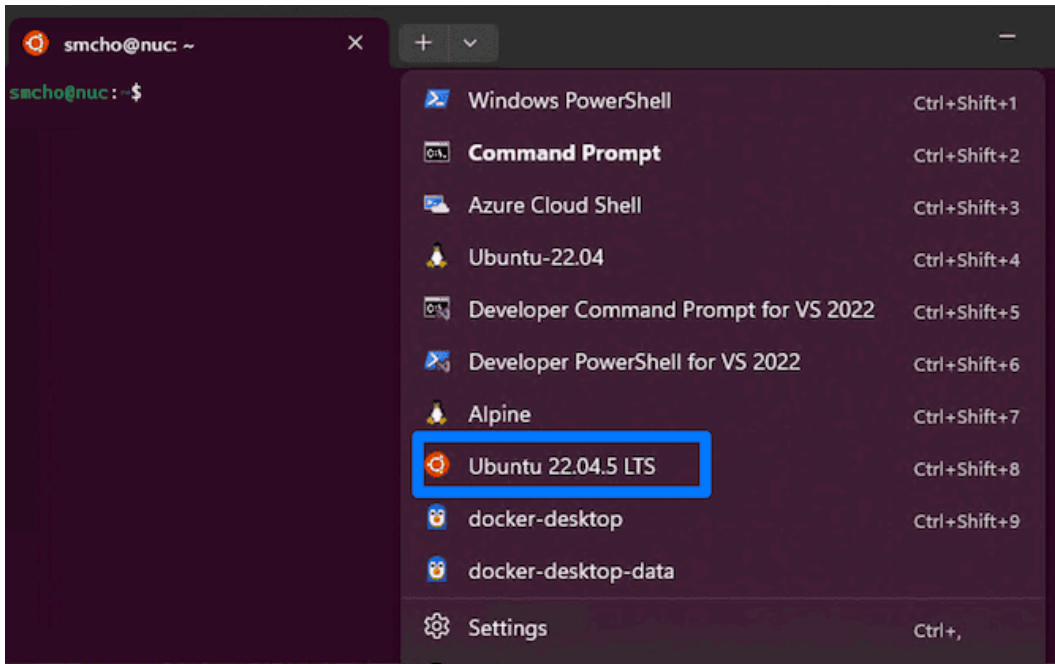
Using WSL2 on Windows

- This page explains how to use **WSL2** (Windows Subsystem for Linux 2).
- All ASE 230 shell scripts, Docker, and Linux command-line tools are written for Linux. **Windows users must run everything inside WSL2.**
- If you are using Linux or macOS, you can skip these slides.

1. Open the WSL2 (Ubuntu) Terminal

You installed WSL2 in HW1.

1. Open the Ubuntu terminal (new tab in Windows Terminal).



2. Do all course work — editing files, running scripts, using Docker —

2. Accessing Windows Files from WSL2

The `C:` drive is mounted as `/mnt/c` inside WSL2.

```
# C:\temp\ex on Windows is the same as:  
cd /mnt/c/temp/ex
```

3. Line-Ending (EOL) Errors

Windows and Linux use different line endings. A script edited on Windows may fail like this:

```
./setup.sh: line 5: $'\r': command not found
```

Fix it with `dos2unix`:

```
sudo apt install dos2unix  
dos2unix ./setup.sh
```

4. Installing Packages for ASE 230 on WSL2

We need PHP 8.3+ on Ubuntu/WSL2.

```
sudo apt update
sudo apt install -y software-properties-common
sudo add-apt-repository ppa:ondrej/php -y
sudo apt update
```

PHP

```
sudo apt install php php-cli php-fpm php-mysql php-zip php-mbstring \
php-xml php-curl php-bcmath php-gd php-intl php-soap unzip curl git -y
```

MySQL

```
sudo apt install mysql-server
sudo service mysql start
sudo service mysql status
```

On a fresh install the server may not be running yet, so start it first.

Use `sudo service mysql start`, not `sudo systemctl start mysql` — most

WSL2 distros don't run systemd by default, and `systemctl` will fail

there ("System has not been booted with systemd"). `service`

Composer

```
curl -sS https://getcomposer.org/installer | php  
sudo mv composer.phar /usr/local/bin/composer  
sudo chmod +x /usr/local/bin/composer  
composer --version
```


Run This Automatically Instead

A `run.sh` is just a script — a text file listing the same commands

you'd otherwise type by hand — that runs them all for you in one step. This one captures everything above (PHP, MySQL, Composer),

plus dos2unix and Composer signature verification. See

`code/02. install_ase230_tools_with_wsl2/readme.txt` for more on

what it does and why.

Navigate to the script's folder first — it has spaces in its name, so quote the path or use Tab-completion:

```
cd "/mnt/c/path/to/your/ase230-repo/ase230/course_tools/code/02. install_ase230_tools_with_wsl2"
# or, if you're already inside the repo you downloaded/cloned:
cd "ase230/course_tools/code/02. install_ase230_tools_with_wsl2"
ls # confirm you see run.sh here
```

Docker (via Docker Desktop, not a separate apt package)

This course uses Docker starting in Module 3. Install it through Docker

Desktop below, which also gives you `docker compose` built in — you do not need to separately `apt install docker-compose`.

Make sure Docker Desktop

(<https://docs.docker.com/desktop/setup/install/windows-install/>) is installed.

- Open Docker Desktop on Windows
- Settings → Resources → WSL Integration
- Turn on "Enable integration with my default WSL distro"
- Turn on integration for your specific distro (Ubuntu, etc.)

Getting Unstuck

Configurations and repositories are updated frequently, so installation

steps may occasionally cause errors. If that happens, you can:

- Copy and paste the error message into an LLM (Claude, ChatGPT, Perplexity, etc.) to get a quick fix.
- Email me the error details.
- Visit me in person so we can troubleshoot together.